

eLandCare User Manual

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1 Introduction

Welcome to eLandCare, your mobile guide to smart soil and water conservation (SWC) in Ethiopia. This application is designed to provide farmers, land managers, and agricultural extension workers with tailored recommendations for SWC practices based on specific local conditions. It also features a tool to explore and identify the Agro-Ecological Zones (AEZ) of your current location.

2 Understanding the SWC Criteria

To get the best recommendations from eLandCare, it's important to understand what each selection criterion means. Here is a detailed guide to the options available in the dropdown menus.

1. Agro-Ecological Zone This refers to the specific climate and altitude zone of your land. The available zones in the app are:

Table 1: Description of agro-ecology zones (MOA, 2016)

GIS value	Name official	Description short (rainfall range and elevation range)
52	Moist High Dega	3200 - 3700 m. a.s.l. and
53	Wet High Dega	3200 - 3700 m. a.s.l. and > 1400mm
41	Dry Dega	2300 - 3000 m. a.s.l. and < 900mm
42	Moist Dega	2300 - 3000 m. a.s.l. and 900 - 1400mm
43	Wet Dega	2300 - 3000 m. a.s.l. and > 1400mm
31	Dry Weyna Dega	1500 - 2300 m. a.s.l. and < 900mm
32	Moist Weyna Dega	1500 - 2300 m. a.s.l. and 900 - 1400mm
33	Wet Weyna Dega	1500 - 2300 m. a.s.l. and > 1400mm
21	Dry Kolla	500 - 1500 m. a.s.l. and < 900mm
22	Moist Kolla	500 - 1500 m. a.s.l. and 900 - 1400mm
23	Wet Kolla	500 - 1500 m. a.s.l. and > 1400mm
11	Dry Berha	Below 500 m. a.s.l. and < 900mm
12	Moist Berha	Below 500 m. a.s.l. and 900 - 1400mm
13	Wet Berha	Below 500 m. a.s.l. and > 1400mm

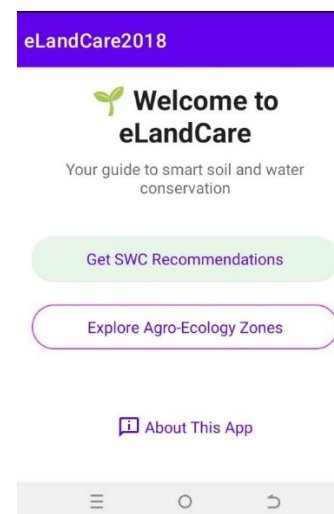
2. Land Use Select how the land is currently being used.
 - Cultivated: Land used for growing crops.
 - Grassland: Land used for animal grazing or left as pasture.

- Forestland: Land covered with trees and forest.
- 3. Slope (% class): This describes the steepness of your land.
 - below 15%: Gently sloping or nearly flat land.
 - 15-50%: Moderately to steeply sloping land.
 - above 50%: Very steep land, such as hillsides.
- 4. Soil Depth (cm): This is the depth of the topsoil, which is crucial for plant growth.
 - above 50 cm: Deep soil, suitable for a wide range of plants.
 - below 50 cm: Shallow soil, which may limit the types of crops that can be grown.
- 5. Soil Type: This refers to the texture of your soil.
 - Sandy/Silty: Lighter soil with larger particles.
 - Clay: Heavy soil with very fine particles.
 - Silt/loam: A balanced mix of sand, silt, and clay, often considered ideal.
- 6. Limiting Factor This is the primary challenge or problem affecting your land.
 - Multiple Gullies: The land is severely affected by many erosion gullies.
 - No Limiting Factor: There are no significant problems currently affecting the land's productivity.
- 7. Drainage: This describes how well water passes through your soil.
 - Well: Water drains at a moderate pace without pooling.
 - Poor: Water drains very slowly, leading to waterlogging.
 - Excess: Water drains too quickly, leaving the soil dry.
- 8. Gully Status This describes the presence and condition of gullies (channels formed by water erosion).
 - No Gully: The land has no significant erosion channels.
 - Gullies present

3 Getting Started

Open the app by clicking the icon  on your mobile. When you first open eLandCare, you will be greeted by the main menu, which provides three primary options:

1. **Get SWC Recommendations:** The core feature of the app. This takes you to a form where you can input land criteria to receive a customized list of conservation practices.
2. **Get Additional Recommendations:** The core feature of the app focusing on gully, road draining, river bank, and spring. This takes you to a form where you can input status of these features to receive a customized list of conservation practices.
3. **Explore Agro-Ecology Zones:** Opens an interactive map to help you identify the AEZ of your location or any other selected point.



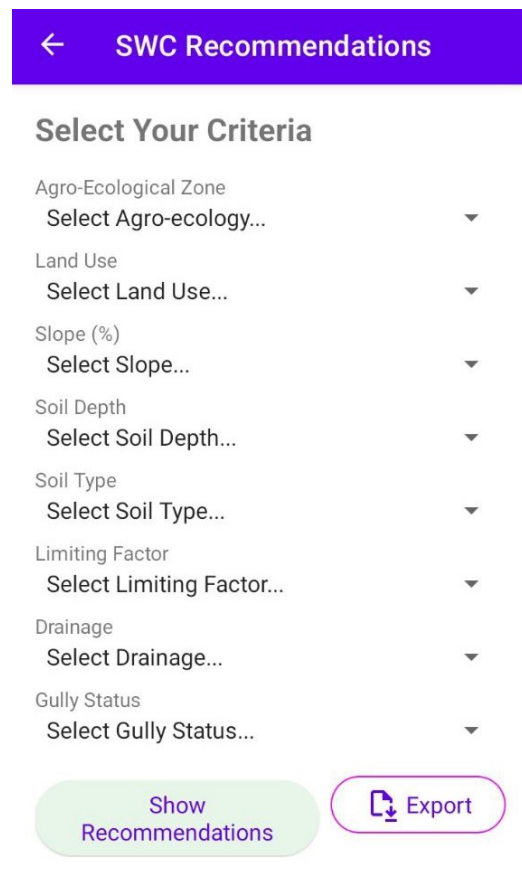
4. **About Conservation Measure:** This takes you to a form where you can get information about the conservation measures: brief description supported by figure/s.
5. **About This App:** Provides information about the application, its purpose, and its developers.

4 How to Get SWC Recommendations

This is the main function of the app. Follow these steps to get your recommendations.

1. **Open the Recommendation Form:** From the main menu, tap the "Get SWC Recommendations" button.
2. **Select Your Criteria:** You will see a form with several dropdown menus. For the most accurate results, select the option that best describes your land for each of the following criteria:

- **Agro-Ecological Zone:** The specific zone where you are standing.
- **Land Use:** The current use of the land (e.g., cropland, grazing land, forest land).
- **Slope class:** The steepness of the land – 0-15%, 15-50%, above 50%
- **Soil Depth class:** The depth of the topsoil: below 50cm, and above 50cm, below or above 50cm.
- **Soil Type:** The primary texture of the soil (e.g., clay, loam, sand, all soil type).
- **Limiting Factor:** The main challenge for the soil (No limiting, Multiple gullies).
- **Drainage:** How well water drains from the soil: well, poor, excess, all class
- **Gully Status:** The presence gullies, no gull present.



The screenshot shows the 'SWC Recommendations' app interface. At the top is a purple header bar with a back arrow and the title 'SWC Recommendations'. Below the header is a section titled 'Select Your Criteria'. This section contains ten dropdown menus, each with a label and a selection prompt, followed by a downward arrow icon. The criteria are: Agro-Ecological Zone (Select Agro-ecology...), Land Use (Select Land Use...), Slope (%) (Select Slope...), Soil Depth (Select Soil Depth...), Soil Type (Select Soil Type...), Limiting Factor (Select Limiting Factor...), Drainage (Select Drainage...), and Gully Status (Select Gully Status...). At the bottom of the form are two buttons: a green button labeled 'Show Recommendations' and a purple button with a download icon labeled 'Export'.

3. **Generate Recommendations:** Once you have made your selections, tap the "Show Recommendations" button.

The following figures illustrate the site information form (left), where users input relevant land and soil data. The right figure displays a completed form with selected

parameters. By clicking the "Show Recommendations" button, users receive tailored bundles of conservation measures for the specified site..

← SWC Recommendations

Select Your Criteria

Agro-Ecological Zone
Select Agro-ecology... ▼

Land Use
Select Land Use... ▼

Slope (%)
Select Slope... ▼

Soil Depth
Select Soil Depth... ▼

Soil Type
Select Soil Type... ▼

Limiting Factor
Select Limiting Factor... ▼

Drainage
Select Drainage... ▼

Gully Status
Select Gully Status... ▼

Show Recommendations

Export

← SWC Recommendations

Select Your Criteria

Agro-Ecological Zone
Dry Kolla ▼

Land Use
Cultivated ▼

Slope (%)
15-50% ▼

Soil Depth
above 50 ▼

Soil Type
All soil type ▼

Limiting Factor
No Factor ▼

Drainage
All class ▼

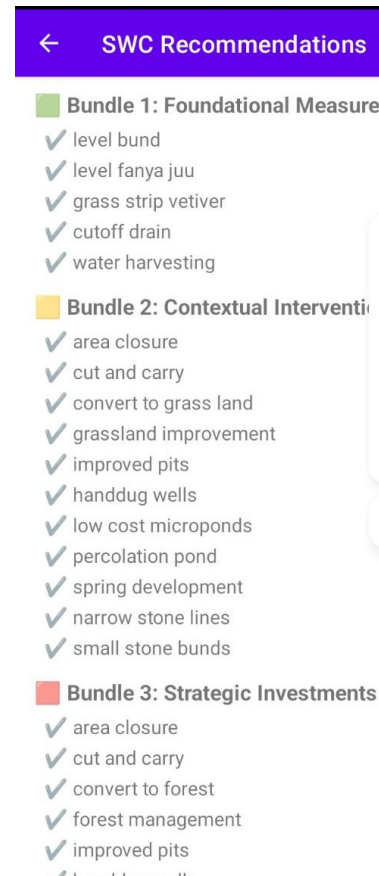
Gully Status
No Gully ▼

Show Recommendations

Export

The next figure displays the soil and water conservation (SWC) recommendations generated for a selected site. These are organized into three bundles—foundational, contextual, and strategic—each tailored to the site's specific conditions and designed to guide effective conservation planning.

1. View the Results: The app will display a list of recommended SWC practices organized into different "Bundles." These bundles represent different levels of investment and intervention, from foundational measures to more advanced techniques. (Suggestion: Replace this with a screenshot of the results)
2. Export Your Results (Optional): If you wish to save your selections and recommendations, tap the "Export" button. This will allow you to save the data as a CSV file (a simple spreadsheet) to your device for later reference or sharing.



5 How to Explore Agro-Ecological Zones (AEZ)

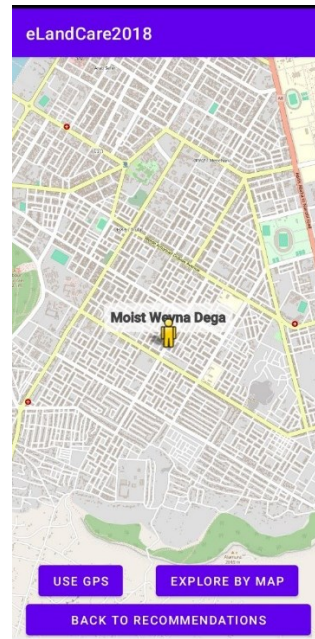
You do not need to use this feature, if you know the agro-ecology of your site. This feature is important for users who has no idea about the agro-ecology of their area of interest. This feature provides two options:

- Generate the agro-ecology of the standing point / the surrounding area (go to 5.1 Explore AEZ of your site in the field)
- Generate the agro-ecology of a place without going to the site by taping on the location on the map (go to [5.2 How to Generate the Agro-ecology of a Site Without Visiting the Field](#))

5.1 Explore AEZ of your site in the field

The AEZ Explorer helps you identify which Agro-Ecological Zone you are in.

3. Open the Map: From the main menu, tap the "Explore Agro-Ecology Zones" button.
4. Grant Location Permission: The app will ask for permission to access your device's location. This is necessary to show your current position on the map. Please grant this permission for the feature to work correctly.
5. Find Your Location: Tap the GPS icon to center the map on your current location. The app will display your current coordinates and the name of the AEZ you are in.



5.2 How to Generate the Agro-ecology of a Site Without Visiting the Field

1. Open the “Explore by Map” Feature
 - From the main screen, tap the “Explore by Map” option.
 - This will open a map, either of the world or Ethiopia, depending on your previous activity.
2. Check Map Availability

If you’ve previously opened the Ethiopia map while connected to the internet, the map data will already be saved on your device.

- In this case, no internet connection is needed.

If the map appears low-resolution or incomplete, you’ll need an internet connection to load detailed data.

3. Zoom to Your Area of Interest

- Use your fingers to navigate (zoom in or zoom out) to the approximate location of your site.
- If detailed map layers don’t appear, connect your mobile device to the internet and zoom in further.

4. Select Your Location

- Once you’ve zoomed in close enough, tap directly on the specific area of interest on the map.
- A pop-up will appear showing:
 - The agroecological zone your site falls within
 - The name of that agroecology

5. Return to the Recommendation Page

After noting the agroecology information, go back to the main Recommendation page.

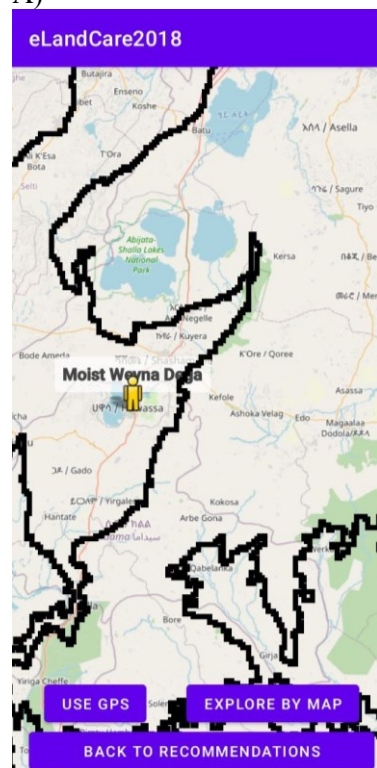
6. Complete the Form and Generate Recommendations

Fill out the required fields in the recommendation form (using the agroecology details you just identified).

Tap “Generate Recommendation” to receive site-specific advice based on recorded data..

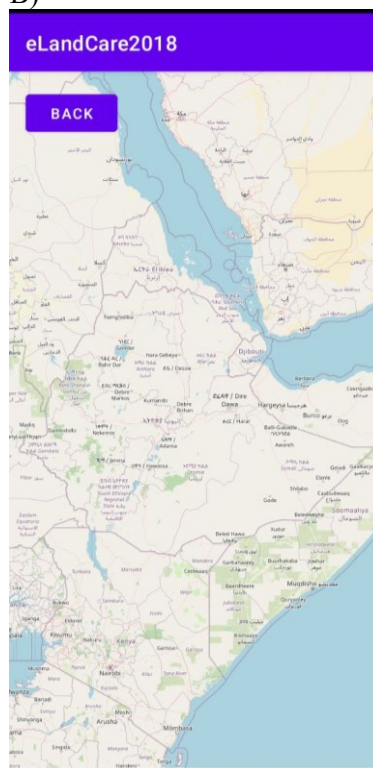
Figure A shows the Agroecological Zone (AEZ) automatically detected when using GPS.

A)



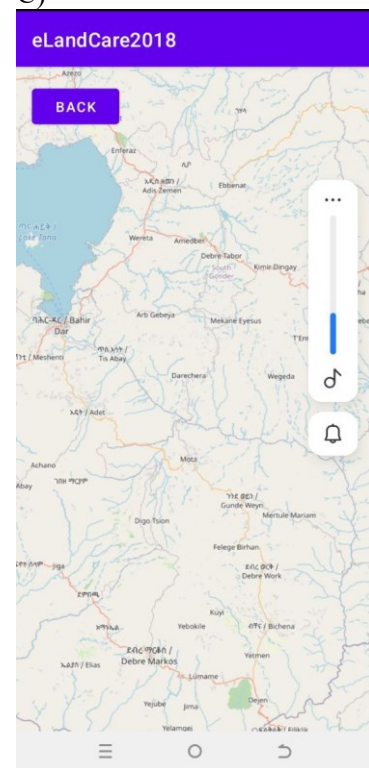
When you instead select the “Explore by Map” option, the app displays a map view as shown in Figure B.

B)

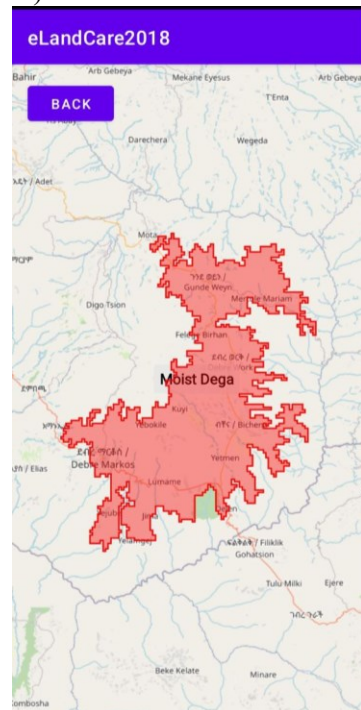


You can then zoom in to your area of interest—Figure C illustrates an example of a user-zoomed location.

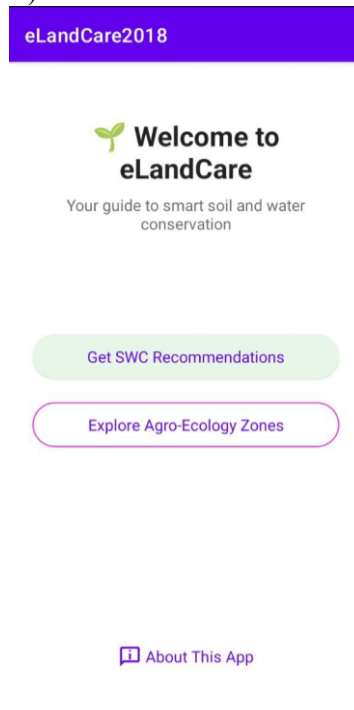
C)



Once you tap on your specific site within the zoomed-in map, a pop-up appears (Figure D) displaying the AEZ boundary and clearly indicating the name of the agroecological zone. D)

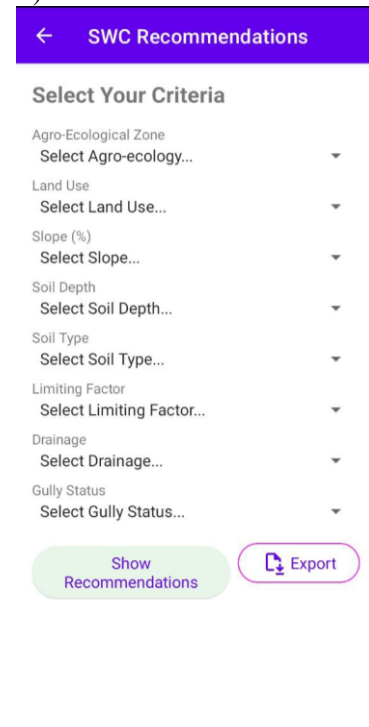


After reviewing this information, tapping the Back button returns you to the main screen (Figure E), from where you can launch the Soil and Water Conservation (SWC) E)



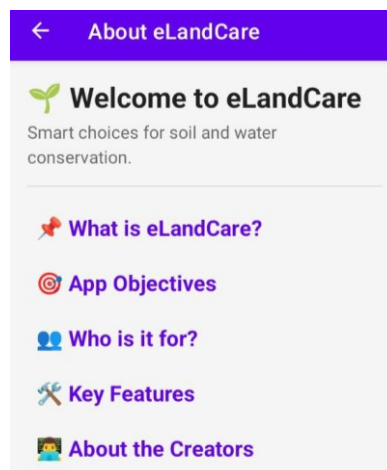
Recommendation page, shown in Figure F, to generate tailored recommendations based on the identified AEZ.

F)



6 About the App

Tapping the "About This App" button will take you to a screen with details about the project, its purpose, acknowledgments, and contact information for support.



7 Frequently Asked Questions (FAQ)

Q: Do I need an internet connection to use the app?

A: The core recommendation feature works completely offline. An internet connection is only needed for the map in the "Explore Agro-Ecology Zones" feature to download map tiles.

Q: Why do I need to grant location permission?

A: Location permission is only used in the "Explore Agro-Ecology Zones" feature to show your current location on the map. It is not used for any other purpose.

Q: The app crashed. What should I do?

A: Please try restarting the app. If the problem persists, please use the contact information in the "About" section to report the issue.

8 How to Install eLandCare

You can install the **eLandCare** application using one of two methods: from the Google Play Store (recommended for most users) or by downloading the installation file directly from GitHub (for advanced users or testing).

8.1 Method 1: Install from the Google Play Store (Recommended)

This is the simplest and most secure way to install the app.

1. Open the Play Store: On your Android device, open the "Google Play Store" application.
2. Search for **eLandCare**: Tap the search bar at the top and type **eLandCare**.
3. Select the App: Find the **eLandCare** app in the search results and tap on it.
4. Install: Tap the "Install" button. The app will automatically download and install on your device.
5. Open: Once installation is complete, you can find the **eLandCare** icon on your home screen or in your app drawer.

8.2 Method 2: Install from GitHub (Sideload)

This method involves manually downloading the .apk installation file from the project's GitHub page. This is useful for installing specific versions of the app that may not be on the Play Store yet.

Step 1: Enable "Install from Unknown Sources" on Your Phone

For security, Android blocks the installation of apps from outside the Play Store by default. You need to grant permission first.

1. Go to your phone's Settings.
2. Navigate to "Apps" or "Security". The exact location varies by Android version.
3. Look for an option named "Install unknown apps" or "Special app access".
4. Find your web browser (e.g., Chrome) in the list and enable the permission to "Allow from this source".

Step 2: Download the APK File from GitHub

1. On your phone's web browser, navigate to the eLandCare GitHub repository's "Releases" page. (releases/ eLandCare-v1.0.apk)
2. Find the latest release and look for the .apk file under the "Assets" section (e.g., eLandCare-v1.0.apk).
3. Tap the .apk file to download it to your device.

Step 3: Install the APK

1. Once the download is complete, open your phone's "Files" or "Downloads" app.
2. Find and tap on the downloaded .apk file (eLandCare-v1.0.apk).
3. A system prompt will appear asking if you want to install the application. Tap "Install".
4. After a few moments, the app will be installed. You can then open it directly or find it on your home screen.

8.3 Method 3: Install from a Computer (via Email or USB Cable)

If you have the eLandCare installation file (eLandCare.apk) on a computer, you can easily transfer it to your phone via email or a USB cable.

8.3.1 How to Install the eLandCare Test Version1.

1. **Enable Installation:** On your phone, go to **Settings > Security** and find the option for **"Install unknown apps"**. Find your browser (e.g., Chrome) or file manager and turn the permission ON. (You only need to do this once.) [if you have a problem with enable installer refer to the different options given below 8.3.2]
2. **Download:** Download the **eLandCare-v1.1-test.apk** file I sent you to your phone.
3. **Install:** Open your "Files" or "Downloads" app, tap on the **eLandCare-v1.1-test.apk** file, and then tap "Install" when prompted.
4. **Test:** The app will now be on your phone. Please use it and report any bugs or crashes

8.3.2 Options to enable installer

Option 1: Use the Search Bar in Settings

Most Android phones have a search bar at the top of the Settings screen.

- Open **Settings**
- Tap the **search icon** () or scroll to the top
- Type: **Install unknown apps** or just **unknown**
- Tap the **Install unknown apps**
- Select your browser (e.g., Chrome) or file manager from the list
- Toggle **Allow from this source** to ON

Option 2: Navigate Manually (if search isn't available)

Depending on your phone brand and Android version, try one of these paths:

- **For Android 8.0 and above:**
- **Settings > Apps & notifications > Advanced > Special app access > Install unknown apps**
- Choose your browser or file manager
- Toggle **Allow from this source**
- **For older Android versions:**
- **Settings > Security > Unknown sources**
- Toggle ON (this applies system-wide)

Option 3: Trigger During Installation

If you're installing an APK directly:

- Tap the APK file
- Android may show a warning and a shortcut to enable permissions
- Tap **Settings** in the popup
- Toggle **Allow from this source**

Option 4: Brand-Specific Paths

Some manufacturers use custom menus:

- **Samsung:** Settings > Biometrics and security > Install unknown apps
- **Tecno/Infinix:** Settings > Apps > Special app access > Install unknown apps

9 Acknowledgment and Developers

9.1 Acknowledgment

The soil and water conservation decision support tool was developed in collaboration with the expertise from the Ministry of Agriculture (MoA) and used many of their resources available as hard copy (e.g. MOA, 2016 agro-ecological classification, soil and water conservation manuals). We are grateful for this. This tool was developed as part of CGIAR science program Scaling for Impact(S4I)in East and Southern Africa, which is grateful for the support of CGIAR Trust Fund contributors (www.cgiar.org/funders) and Nature-Based Solutions for Inclusive and Sustainable Development (NSSID) funded by Swedish SIDA.

9.2 Developed

- Initiated and conceptualized by Amare Hailelassie, Principal Researcher, Agricultural Water Management and Ecosystem Services, International Water Management Institute (IWMI), and Wolde Mekuria, Senior Researcher, Environment nd Development, IWMI
- Developed by: Kefyalew Sahele – GIS & Mobile Application Developer; Expertise in Forestry and Land Use Planning

9.3 Validated

- Fekede Adane and Abinet Mengistu, soil and water conservation expert, Ministry of Agriculture, and Tarku Temesegen, rural development, Farm Africa